



SwarmOps automates the hardest part of operating a drone fleet: deciding, in real time, which drone flies which mission, in what order and along which route — while guaranteeing no drone runs out of battery mid-flight. A working distributed system, deployed on AWS and demonstrable live today.

Optimization (VRPTW)

Optimal assignment and routing, proven battery feasibility on every route

Re-planning in seconds

Automatic response to real-time disruptions, no manual trigger

Deployed on AWS

Secure, distributed and scalable cloud infrastructure

Working PoC

Built and running; live demonstration available on request

The Problem

Beyond a handful of drones, fleet coordination is no longer solvable by a human dispatcher: battery limits that depend on both distance and payload weight, priorities and deadlines, no-fly zones, charging logistics and mid-mission disruptions — all interacting at once. The result: wasted flight time, duplicate coverage, missed deadlines, stranded aircraft and an inaccurate operational picture. Manufacturer software flies one aircraft well; nobody answers the fleet-level question — who should do what, and can they actually complete it?

WHY WE BUILT THIS

This is not a theoretical scenario for us. On 7 October 2023, during the fighting in the kibbutzim, two teams from the same company launched drones at the same time. With no coordination between them, both swept the same sector — one team identified hostiles, the other reported a friendly force in the same place. The contradiction in the situational picture delayed a critical operational decision. That is the problem SwarmOps was built to solve.

The Product — Built, Not Planned

- **Optimal assignment and routing** — measured against a greedy baseline; proven battery feasibility on every route, with charging stops inserted automatically.
- **Real-time re-planning** — live telemetry streams through a message bus; a disruption re-solves the affected schedule within seconds.
- **Live control room** — real-time map, mission board, fleet inventory, operational dashboards.
- **Enterprise-grade infrastructure** — secure, distributed and scalable cloud, with automated and controlled deployments; the maturity startups build in year two, already in place.

Current scope, stated plainly: the fleet is simulated — a deliberate scope decision. Connecting real aircraft (MAVLink protocol) is Phase 1 engineering, not research risk.

Roadmap — 12 Months

Months 0–3

Real aircraft: incorporate, MAVLink, full loop on hardware, CAAI groundwork.

Months 3–6

First pilot: design partner, success criteria agreed up front.

Months 6–12

Productization: Multi-Tenant, pricing, pilot→paying customer, Pre-Seed / Seed round.

Market & Use Cases

Segment	Use case
Defense & security	Resupply tasking, patrol/surveillance coverage, inspection rounds
Infrastructure & energy	Power lines, pipelines, solar fields — recurring missions
Agriculture	Survey and spraying coordination at scale
Delivery	Last-mile / site-to-site (regulation-dependent)

Additional defense capabilities in development (Phase 2–3)

- **Fleet monitoring and predictive maintenance** — maintenance status already exists as a core field; predictive scoring will pull a drone from the available pool before it fails.
- **Single command picture across units** — a unified map for a single fleet already exists; multi-unit hierarchy (unit → base → command) is in development.
- **Eliminating manual dispatch inefficiency** — optimized assignment and routing, proven against a manual/greedy baseline, are built and measured.

Positioning: a vendor-neutral coordination layer above whatever aircraft an organization already owns. Fleets are scaling, BVLOS regulation is opening, and the Israeli ecosystem is a natural first market.

Team

Tony [surname] (technical lead) · Guy [surname] · Harel [surname] — three engineers who designed, built and operate the entire system: algorithms, services, frontend, cloud infrastructure and delivery pipeline.

Field and combat unit veterans; two of us are drone operators — operational understanding from the field, combined with advanced software engineering.

The Ask

Incubation / accelerator — workspace and program affiliation · Mentorship (business development, defense/aerospace) · Introductions (operators, integrators, infrastructure companies) · Pre-Seed support ([amount]: development drones, cloud, legal/regulatory). Live demonstration available on request.